

1 Mammals have a closed double circulation system.

(a) Explain what is meant by a *closed double circulation*.

.....

.....

.....

.....

.....[2]

(b) Table 1.1 shows some structures in the mammalian circulatory system.

Complete Table 1.1 to show the sequence of structures through which blood flows, starting with the pulmonary vein.

Use the numbers 2 to 5 to indicate the correct sequence.

Table 1.1

structure	sequence of blood flow
left ventricle	
vena cava	
pulmonary vein	1
aorta	
right atrium	

[2]

(c) (i) Explain why arteries have thicker walls than veins.

.....

.....

.....

.....

.....[2]

(ii) Smoking causes carbon monoxide and nicotine to enter the blood.

Describe the short-term effects of each of these substances on the cardiovascular system.

carbon monoxide

.....

.....

.....

nicotine

.....

.....

.....[4]

[Total: 10]